Target-Conditioned Evidence Retrieval for Knowledge Tracing

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Abstract. Knowledge tracing predicts student responses from their learning histories to support personalized learning. As intelligent tutoring systems (ITS) expand their question banks, they must predict responses to questions involving new knowledge components (KCs). Historical responses can inform these predictions through shared concepts, cognitive demands, and structural relations. Evidence selection therefore needs to consider both target relevance and complementary information. We propose Cognitive-RAG KT, a target-conditioned evidence retrieval framework. It recalls candidate interactions through multiple relations and selects a complementary evidence set by balancing target relevance, relation coverage, and redundancy. Structured relation attributes and a schema-constrained summary organize the selected responses into a prediction context. This context and the target-question representation are projected into continuous soft slots of a frozen language model to predict the probability of a correct response. Experiments on three public datasets show that Cognitive-RAG KT outperforms the evaluated KT baselines.

Keywords: Knowledge Tracing · Educational Data Mining · Large Language Models

1 Introduction

Intelligent tutoring systems (ITS) support personalized practice and learning assistance. Knowledge tracing (KT) uses students' learning histories to predict whether they will answer questions correctly, providing a basis for these services. As question banks expand, students encounter questions involving new knowledge components (KCs). Predicting responses to these questions requires connecting the demands of new content with students' performance in previous practice.

Recurrent and memory-based models, including DKT and DKVMN, encode historical responses into an evolving representation of a student's knowledge [18, 24]. Attention-based methods such as SAKT and AKT assign attention weights to historical interactions according to their relevance to the target question [17, 7]. Retrieval-based KT further incorporates selected records into prediction, as illustrated by CoKT's retrieval of response sequences from similar students [16]. For new questions and concepts, inductive approaches use textual descriptions and structural relations to construct representations beyond question and concept identifiers [6]. These approaches represent educational content and relate past responses to subsequent performance.

Textual and structural representations provide a basis for connecting questions involving new KCs to previously observed content. We focus on selecting historical responses

that jointly inform prediction for such questions. Historical questions may connect to the target through shared concepts, prerequisite knowledge, or similar cognitive demands. Selecting several highly similar questions may repeat information about one requirement while leaving others poorly represented. Evidence selection should therefore account for both the relevance of each interaction and the additional information it contributes to the selected set.

Consider a question that asks for the final bead's color and the number of beads of that color in a repeating arrangement. Answering it involves both locating an element within a cycle and counting occurrences across cycles. Responses to historical position-finding and counting questions reflect performance on these two requirements, respectively. Selecting several position-finding questions may yield repeated information without showing the student's performance on counting. Combining responses from both types provides evidence about both requirements of the target.

To select such evidence, we propose Cognitive-RAG KT, a target-conditioned evidence retrieval framework for predicting responses to questions involving new KCs. Candidate recall uses conceptual, cognitive, graph, and collaborative relations to identify historical interactions connected to the target. Reranking and greedy selection balance target relevance, relation coverage, and redundancy to construct an evidence set. Each selected response retains attributes describing its question's relation to the target. A schema-constrained summary brings these observations together into an account of the student's relevant performance. The evidence context and target-question representation are then projected into soft slots of a frozen language model to predict the probability of a correct response.

This work makes two contributions to predicting student responses on questions involving new KCs. First, we develop a two-stage retrieval strategy that selects complementary historical interactions by balancing target relevance, incremental concept and relation coverage, and redundancy within the evidence set. Second, we construct an evidence context from selected responses, their relations to the target, and a structured summary, and connect it to a frozen language model through soft slots. Experiments on three public datasets show improved prediction performance, while ablation studies assess the contributions of evidence selection and context construction.

2 Related Work

2.1 Modeling Student Interaction Histories

State-based KT models describe student knowledge through probabilistic transitions or learned representations of response sequences. Bayesian Knowledge Tracing models mastery of each KC as a binary state that evolves through probabilistic transitions [3]. DKT uses a recurrent neural network to learn a sequential representation of student knowledge from past responses [18]. DKVMN organizes concepts in a static key memory and updates student knowledge in a dynamic value memory [24].

Attention-based KT models weight historical interactions by their relevance to the target question. SAKT identifies relevant past interactions through self-attention to predict a student's response [17]. In AKT, attention weights combine question similarity

with context-aware distance and exponential decay [7]. simpleKT represents differences among questions through question-specific variations and applies ordinary dot-product attention to historical interactions [14]. sparseKT retains high-scoring historical interactions through soft-thresholding or top-K sparse attention [10]. DenoiseKT combines graph-enhanced question representations with denoised attention to reduce the influence of irrelevant historical information [4].

2.2 Inductive Representations for Questions and Knowledge Components

Inductive KT uses textual and structural information to represent newly introduced questions and concepts. SINKT encodes question and concept descriptions with a pretrained language model to obtain semantic representations [6]. It constructs a heterogeneous graph from question-concept associations and LLM-generated concept relations, then aggregates neighboring representations to incorporate structural information into response prediction.

2.3 Retrieval and Language Models in Knowledge Tracing

Retrieval-based KT draws on historical records and structured resources to construct context for prediction. CoKT retrieves response records and sequences from students with similar learning histories and integrates this inter-student information with the target student's history [16]. RAG-KT combines behavioral, structural, and ability-based similarities to retrieve similar students and organizes individual and peer performance into a structured context for cross-platform prediction and interpretable diagnosis [5]. Our retrieval selects interactions from the target student's own history, using target relevance together with concept and relation coverage gains and redundancy relative to the selected evidence.

Language-model-based KT combines question semantics with student interaction representations for response prediction. LLM-KT uses plug-in context and sequence components to map textual and behavioral representations into a language model's embedding space [21]. Related parameter-efficient adaptation methods use trainable continuous representations to adapt frozen language models to downstream tasks. Prefix tuning learns prefix activations across Transformer layers, whereas prompt tuning prepends trainable embeddings to the model input [12, 11].

In Cognitive-RAG KT, history soft slots are constructed from selected response records, their relations to the target, and a structured summary. A separate projector maps the target-question representation into target soft slots.

3 Method

3.1 Problem Formulation

Let $\mathcal{U}$, $\mathcal{Q}$, and $\mathcal{C}$ denote the sets of students, questions, and knowledge components (KCs), respectively. Each question $q \in \mathcal{Q}$ is associated with a concept set $\mathcal{C}(q) \subseteq \mathcal{C}$. Let $\mathcal{C}_{\text{tr}}$ denote the training concepts, which occur in the training response records. The new concepts $\mathcal{C}_{\text{ho}}$ do not occur in these records, with $\mathcal{C}_{\text{tr}} \cap \mathcal{C}_{\text{ho}} = \emptyset$.

For a student u, the response history before step t is

$$\mathcal{H}_t = \{(q_i, y_i) \mid 1 \le i < t,\ C(q_i) \subseteq C_{\text{tr}}\},$$

where q_i is the question attempted at step i and $y_i \in \{0, 1\}$ is its observed response outcome. Given $\mathcal{H}_t$ and a target question q_t, the model estimates the probability of a correct response:

$$\hat{p}_t = P(y_t = 1 \mid \mathcal{H}_t, q_t).$$

Test targets satisfy $C(q_t) \cap C_{\text{ho}} \neq \emptyset$.

3.2 Framework Overview

Figure 1 summarizes Cognitive-RAG KT. Given a student's interaction history $\mathcal{H}_t$ and a target question q_t, the framework has four components: question representation, historical interaction retrieval, context construction, and response prediction. First, it represents the target and historical questions using textual, semantic, and collaborative information. Second, it scores historical interactions by their relations to q_t, reranks the retained candidates, and selects an evidence set E_t containing up to K interactions from $\mathcal{H}_t$. Third, it encodes the selected interactions, their response outcomes, and a structured summary into a context representation. Finally, it projects the context and target-question representations into the embedding space of a frozen large language model. The correct response probability $\hat{p}_t$ is computed from the logits of the correct and incorrect label tokens.

3.3 Question Representation

For each question q, we construct text, hierarchical semantic, semantic, and collaborative representations. The hierarchical semantic representation encodes representative words from coarse- and fine-grained clusters. The semantic representation fuses text and hierarchical semantics and adjusts the result using cognitive-level information and Rasch difficulty. The four representations are used in historical interaction retrieval and question context construction.

Text and Hierarchical Semantic Representations. Let x_q denote the textual content of question q. A fixed pretrained text encoder maps x_q to $\mathbf{h}_q^{\text{text}}$.

Hierarchical semantics extraction.

We apply two-level clustering to the text representations of training questions. The first level forms coarse-grained clusters, and the second partitions each coarse-grained cluster into fine-grained clusters.

Hierarchical semantics translation.

We describe each question's cluster assignments using representative words: the most frequent word from its coarse-grained cluster and up to two frequent valid words from its fine-grained cluster. These words provide general and specific semantic descriptions, respectively. The resulting word sequence is encoded by the same text encoder to obtain $\mathbf{h}_q^{\text{hier}}$.

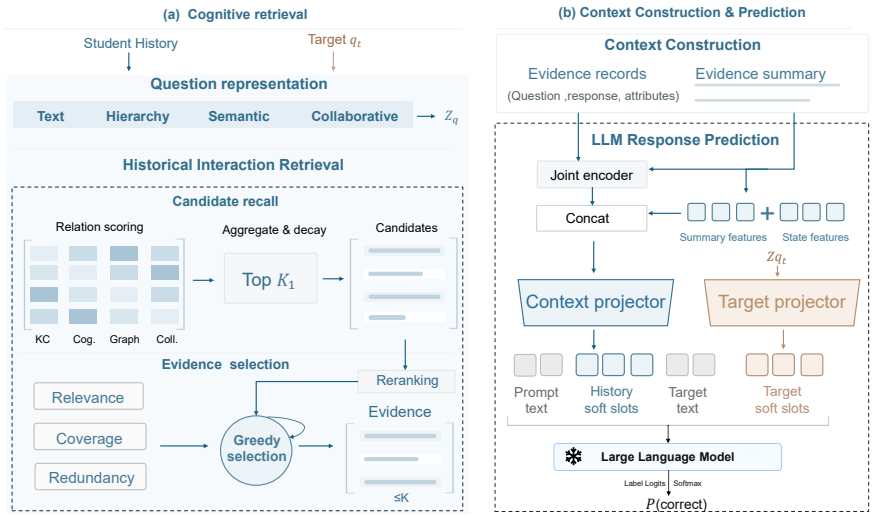


Fig. 1. Overview of Cognitive-RAG KT. (a) Target-conditioned retrieval recalls historical interactions using multiple relations and selects up to K complementary evidence interactions through reranking and greedy selection. (b) Evidence records and a schema-constrained summary are jointly encoded and concatenated with summary and state features. The resulting context and target-question representation are mapped into continuous soft slots to condition a frozen LLM for response prediction.

Semantic Representation. The text representation and the hierarchical semantic representation are concatenated and mapped through an MLP to obtain $\mathbf{e}_q = \mathrm{MLP}([\mathbf{h}_q^{\mathrm{text}}; \mathbf{h}_q^{\mathrm{hier}}])$.

For each concept c, let $\mathcal{Q}_c$ denote its associated questions. We apply PCA to the representations $\{\mathbf{e}_q \mid q \in \mathcal{Q}_c\}$ and take the first principal component as the concept direction. We orient this direction using the cognitive-level annotations of these questions, so that larger projections correspond to higher cognitive levels. The resulting unit vector is denoted by $\mathbf{d}_c$.

For a question associated with multiple concepts, the corresponding concept directions are averaged and normalized to obtain $\mathbf{d}_q$. The Rasch model provides the question-difficulty parameter b_q [19], which is mapped to the bounded coefficient $\mu_q = \tanh(ab_q)$. The semantic representation of question q is defined as

$$\mathbf{h}_q^{\mathrm{sem}} = \mathbf{e}_q + \mu_q \mathbf{d}_q. \tag{1}$$

Here, $\mathbf{d}_q$ determines the direction of the adjustment, while μ_q determines its signed magnitude. We aggregate only available concept directions; if none is available or their mean has zero norm, we set $\mathbf{d}_q = \mathbf{0}$, yielding $\mathbf{h}_q^{\mathrm{sem}} = \mathbf{e}_q$.

Collaborative Representation. We train a skip-gram Word2Vec model on chronologically ordered question sequences constructed from training interactions. The learned

embedding $\mathbf{h}_q^{\text{col}}$ captures which questions tend to appear near question q in these sequences.

Final Question Representation. The four representations are concatenated as

$$\mathbf{z}_q = \left[\mathbf{h}_q^{\text{text}}; \mathbf{h}_q^{\text{hier}}; \mathbf{h}_q^{\text{sem}}; \mathbf{h}_q^{\text{col}}\right]. \tag{2}$$

For the target question q_t, $\mathbf{z}_{q_t}$ is used for prediction.

3.4 Historical Interaction Retrieval

Given the student interaction history $\mathcal{H}_t$ and the target question q_t, the retrieval module first recalls potentially relevant historical interactions and then selects a compact evidence set. The resulting evidence set E_t is used to construct the historical evidence context for predicting the response to q_t.

Candidate Recall. For each admissible interaction, write $h_i = (q_i, y_i)$, $C_i = C(q_i)$, and $C_t = C(q_t)$. For each historical interaction $(q_i, y_i) \in \mathcal{H}_t$, we compute a recall score according to its relation to the target question q_t and its position in the interaction history. The relation is evaluated using concept similarity, cognitive levels, graph relations, and collaborative similarity.

Let $J_{i,t}$ be the Jaccard similarity between C_i and C_t. Define the cognitive-level difference as $\Delta\ell_{i,t} = \ell_{q_t} - \ell_{q_i}$ and the semantic similarity as $s_{i,t} = \cos(\mathbf{h}_{q_i}^{\text{sem}}, \mathbf{h}_{q_t}^{\text{sem}})$. For a lower-level historical question, we combine its observed response with a diagnostic probability:

$$M_{i,t} = \rho y_i + (1-\rho) p_{i|t}^{\text{diag}}, \tag{3}$$

where $p_{i|t}^{\text{diag}}$ estimates the probability of answering q_i correctly under a target-conditioned student state constructed from recent admissible interactions, and ρ controls the balance between the observed response and the diagnostic estimate. The cognitive contribution is

$$L_{i,t} = \begin{cases} w_{\text{pre}} J_{i,t} M_{i,t} e^{-\gamma_{\text{pre}} \Delta\ell_{i,t}}, & \Delta\ell_{i,t} > 0, \\ w_{\text{peer}} s_{i,t}, & \Delta\ell_{i,t} = 0,\ J_{i,t} + G_{i,t} > 0, \\ w_{\text{semantic}} s_{i,t}, & \Delta\ell_{i,t} = 0,\ J_{i,t} + G_{i,t} = 0, \\ w_{\text{high}} J_{i,t} [\beta_+ y_i - \beta_-(1-y_i)] e^{-\gamma_{\text{high}} |\Delta\ell_{i,t}|}, & \Delta\ell_{i,t} < 0. \end{cases} \tag{4}$$

Here $G_{i,t}$ is the structural contribution defined below. The branch weights control the contributions of different cognitive relations. The γ coefficients govern attenuation with cognitive-level distance, and β_+ and β_- weight correct and incorrect higher-level responses, respectively. For questions at the same cognitive level, an explicit concept or graph match defines the peer relation; otherwise, semantic similarity defines the semantic relation.

The graph contribution $G_{i,t}$ captures structural connections beyond direct concept overlap. It is the maximum weighted prerequisite or neighbor match over concept pairs from C_i and C_t, with zero assigned when no match exists. This contribution allows historical questions involving related concepts to support target prediction.

Collaborative similarity provides an additional relation derived from the collaborative representations of q_i and q_t. It is downweighted when the concept, cognitive level, and graph relations already provide strong evidence. The resulting collaborative score is denoted by $C_{i,t}$.

The four relations are combined as

$$B_{i,t} = w_J J_{i,t} + L_{i,t} + w_{\text{graph}} G_{i,t} + w_{\text{col}} C_{i,t}, \tag{5}$$

where $L_{i,t}$ includes the weight for the cognitive level relation. The remaining coefficients weight concept similarity, graph relations, and collaborative similarity.

We further account for the cumulative semantic difference between the historical interaction and the target position:

$$D_{i,t} = 1 + \sum_{\substack{i<j<t \\ (q_j, y_j) \in \mathcal{H}_t}} \left[1 - \text{sim}\left(\mathbf{h}^{\text{sem}}_{q_j}, \mathbf{h}^{\text{sem}}_{q_t}\right)\right]. \tag{6}$$

Here, $\text{sim}(\cdot, \cdot)$ denotes cosine similarity. The sum includes only admissible historical interactions between q_i and q_t. A larger $D_{i,t}$ indicates greater cumulative semantic difference from the target question.

The final recall score is

$$R_{i,t} = \exp\left(-\alpha D_{i,t}\right) B_{i,t}, \tag{7}$$

where α controls the effect of cumulative semantic difference on the recall score. The K_1 historical interactions with the highest recall scores form the candidate set $\mathcal{A}_t$.

Evidence Selection. Candidate recall reduces the interaction history to a candidate set $\mathcal{A}_t$. Selecting only the highest scoring candidates may retain interactions that cover the same concepts or share the same relation to the target. We therefore select evidence according to target relevance, coverage of concepts or relations absent from the selected set, and similarity to interactions already selected.

For each recalled candidate, selection uses the relation score $B_{i,t}$ without semantic decay and adds a reranker contribution:

$$U_{i,t} = B_{i,t} + w_{\text{rr}} r^{\text{rr}}_{i,t}, \tag{8}$$

where $r^{\text{rr}}_{i,t}$ is the relevance probability obtained from the reranker's yes/no label logits.

Let $\mathcal{S} \subseteq \mathcal{A}_t$ denote the evidence selected so far. Candidate recall retains up to K_1 interactions. Evidence selection constructs a final set with a budget of $K_2 = K \leq K_1$ interactions. The actual number selected for target q_t is $K_t = \min(K, |\mathcal{A}_t|)$. To construct complementary evidence, we reward newly covered relations and concepts while penalizing similarity to selected interactions. We distinguish six target-conditioned relation

roles: prerequisite-level, peer-level, semantic, higher-level, graph, and collaborative. They are denoted by pre, peer, semantic, high, graph, and collab, respectively. The active branch of $L_{i,t}$ supplies the corresponding cognitive-role score, while inactive cognitive roles receive zero. The graph and collaborative role scores are $S_{i,t}^{\text{graph}} = G_{i,t}$ and $S_{i,t}^{\text{collab}} = C_{i,t}$. Let $\mathcal{R}_i = \{r : S_{i,t}^{r} > \tau_r\}$ denote the active relation roles of h_i with respect to q_t, where τ_r is the activation threshold for role r. Define $O_i = C_i \cap C_t$ and $\mathcal{V}_i = \mathcal{N}(q_i) \cap \mathcal{N}(q_t)$, where $\mathcal{N}(q)$ is the stored set of neighboring concepts. For selected evidence $\mathcal{S}$, let $\mathcal{R}_{\mathcal{S}}$, $O_{\mathcal{S}}$, and $\mathcal{V}_{\mathcal{S}}$ denote the corresponding unions. Coverage gain is

$$\begin{aligned}\text{Cov}(h_i \mid \mathcal{S}) = |\mathcal{R}_i \setminus \mathcal{R}_{\mathcal{S}}| \\ + \frac{|O_i \setminus O_{\mathcal{S}}|}{|C_t| + \varepsilon} + \frac{|\mathcal{V}_i \setminus \mathcal{V}_{\mathcal{S}}|}{|\mathcal{N}(q_t)| + \varepsilon},\end{aligned} \tag{9}$$

where $\varepsilon > 0$ ensures numerical stability. Redundancy combines concept overlap, overlap between target-conditioned relation roles, and semantic similarity:

$$\text{Red}(h_i, \mathcal{S}) = \max\left(\{0\} \cup \{0.4J_{i,j} + 0.2J_{i,j}^{\mathcal{R}} + 0.4s_{i,j} : h_j \in \mathcal{S}\}\right), \tag{10}$$

where $J_{i,j}$ is concept Jaccard similarity, $J_{i,j}^{\mathcal{R}} = |\mathcal{R}_i \cap \mathcal{R}_j|/(|\mathcal{R}_i \cup \mathcal{R}_j| + \varepsilon)$, and $s_{i,j}$ is semantic cosine similarity. Thus, a candidate is penalized according to its strongest similarity to existing evidence.

The selection score combines target relevance, coverage gain, and redundancy:

$$F(h_i \mid \mathcal{S}) = U_{i,t} + \lambda_{\text{cov}} \text{Cov}(h_i \mid \mathcal{S}) - \lambda_{\text{red}} \text{Red}(h_i, \mathcal{S}), \tag{11}$$

where λ_{cov} and λ_{red} control the coverage reward and redundancy penalty, respectively.

We initialize the evidence set with the interaction most relevant to the target:

$$h^{(1)} = \arg\max_{h_i \in \mathcal{A}_t} U_{i,t}. \tag{12}$$

At each subsequent step $k = 2, \ldots, K_t$, we select the candidate with the highest $F(h_i \mid \mathcal{S}_{k-1})$. This score rewards target relevance and additional coverage while penalizing redundancy with the selected evidence:

$$h^{(k)} = \arg\max_{h_i \in \mathcal{A}_t \setminus \mathcal{S}_{k-1}} F(h_i \mid \mathcal{S}_{k-1}), \qquad \mathcal{S}_k = \mathcal{S}_{k-1} \cup \{h^{(k)}\}, \tag{13}$$

where $\mathcal{S}_1 = \{h^{(1)}\}$. After K_t interactions have been selected, the final evidence set is

$$E_t = \mathcal{S}_{K_t}. \tag{14}$$

3.5 Context Construction

Each selected interaction is represented by its question text, observed response, relation role, overlapping concepts, cognitive-level difference, and selection support score. The

support score is the score used at the step when the interaction was selected. We serialize these records in greedy selection order to form the evidence text T_t^{evidence}.

A schema-constrained summary provides a compact account of the selected evidence, including mastered and weak concepts, transfer state, response risk, evidence quality, and a concise diagnosis. The summary generator receives target information, serialized evidence attributes, and a performance summary. The performance summary is computed from the selected evidence and the dynamic student prior. We jointly encode the rendered summary T_t^{summary} and evidence text, then concatenate the encoding with summary features $\mathbf{f}_t$ and state features $\mathbf{n}_t$:

$$x_t^{\text{hist}} = [\text{Enc}(T_t^{\text{summary}} \| T_t^{\text{evidence}}); \mathbf{f}_t; \mathbf{n}_t], \qquad x_t^{\text{ques}} = \mathbf{z}_{q_t}. \tag{15}$$

Here $\mathbf{f}_t$ contains summary features derived from concept counts, concept-presence indicators, diagnosis length, risk, and evidence quality. The state features $\mathbf{n}_t$ contain the dynamic student prior and normalized candidate and evidence counts. Target text, concept annotations, and cognitive level also provide textual information in the prediction prompt. The historical and target contexts jointly condition the response predictor.

3.6 LLM Response Prediction

The prediction module maps the historical context x_t^{hist} and the question context x_t^{ques} into the token embedding space of the frozen LLM. The LLM then estimates the probability of a correct response from the logits of two candidate labels.

Context Projection. Two projection networks map the historical and question contexts into the frozen LLM's token embedding space:

$$\mathbf{Z}_t^{\text{hist}} = P_\phi^{\text{hist}}\left(x_t^{\text{hist}}\right), \qquad \mathbf{Z}_t^{\text{ques}} = P_\psi^{\text{ques}}\left(x_t^{\text{ques}}\right), \tag{16}$$

where $\mathbf{Z}_t^{\text{hist}}$ and $\mathbf{Z}_t^{\text{ques}}$ are sequences of continuous embeddings compatible with the token embedding dimension of the LLM.

The historical context embeddings represent the selected evidence, its attributes, and the schema-constrained summary. The question embeddings represent the current target question. Both groups are incorporated into a fixed knowledge tracing prompt to condition response prediction.

Label-Logit Prediction. The prompt defines the knowledge tracing task and two candidate labels for correct and incorrect responses. Given the prompt and the two groups of continuous embeddings, the LLM produces a logit for each label.

Let o_t^+ and o_t^- denote the logits assigned to the correct and incorrect labels, respectively. The predicted probability of a correct response is computed as

$$\hat{p}_t = \frac{\exp(o_t^+)}{\exp(o_t^+) + \exp(o_t^-)}. \tag{17}$$

Given the observed response $y_t \in \{0, 1\}$, the prediction loss is defined as

Table 2. Statistics of training and test targets and KCs. Percentages indicate the proportion of new KCs among all KCs appearing in test targets.

Dataset	Train targets	Test targets	Train KCs	Test KCs	New KCs
FoundationalASSIST	328,433	666,034	27	127	113 (89.0%)
XES3G5M	108,319	496,715	99	524	476 (90.8%)
MOOCRadar	66,512	818,197	138	2,703	2,642 (97.7%)

$$\mathcal{L}_{\text{pred}} = -\sum_{t} \left[y_t \log \hat{p}_t + (1 - y_t) \log(1 - \hat{p}_t) \right]. \tag{18}$$

Before response-prediction training, we train the question-representation MLP and the model producing the dynamic diagnostic prior, then fix their parameters. Response-prediction training updates only the historical-context and question-context projectors. The precomputed feature generators and the prediction LLM also remain fixed.

4 Experimental Setup

We evaluate Cognitive-RAG KT on response prediction for questions involving new KCs, compare it with standard KT models, and examine the contributions of retrieval and evidence construction.

4.1 Datasets and Evaluation Protocol

We use three public KT datasets: FoundationalASSIST [22], XES3G5M [15], and MOOCRadar [23]. Each dataset contains chronologically ordered student–question interactions with binary response outcomes and question–concept annotations. Table 1 summarizes the datasets before evaluation targets are constructed.

Table 1. Statistics of the three datasets before target construction.

Dataset	Students	Questions	Concepts	Interactions
FoundationalASSIST	5,000	2,492	143	999,467
XES3G5M	4,833	3,903	575	609,867
MOOCRadar	14,224	9,383	9,985	898,933

A question $q_{u,t}$ is included as a test target when $C(q_{u,t}) \cap C_{\text{ho}} \neq \emptyset$.

For each target, the student history is

$$\mathcal{H}^{\text{tr}}_{u,<t} = \left\{ (q_{u,i}, y_{u,i}) \mid i < t,\ C(q_{u,i}) \subseteq C_{\text{tr}} \right\}.$$

The model uses the questions and observed responses in this history, together with target information such as question text and concept annotations.

We exclude each student's first interaction because it has no preceding history. Table 2 reports the resulting training and test target counts and concept statistics.

4.2 Baselines

We compare Cognitive-RAG KT with twelve KT baselines. DKT [18] uses a recurrent neural network to encode response histories. SAKT [17] uses self-attention to relate historical interactions to the target question. SAINT [2] encodes question and response sequences with a Transformer encoder–decoder. AKT [7] combines question similarity with context-aware distance and exponential decay in its attention weights. simpleKT [14] combines question-specific variations with ordinary dot-product attention.

DIMKT [20] incorporates question difficulty into question representations and knowledge-state updates. QIKT [1] models question-sensitive knowledge acquisition and problem solving and uses an item response theory prediction layer. KeenKT [13] represents knowledge states with normal inverse Gaussian distributions and uses distribution-based attention to track their evolution.

sparseKT [10] uses sparse attention to retain high-scoring historical interactions. DenoiseKT [4] enhances question representations with a graph neural network and suppresses irrelevant interactions through denoised attention. TCKT [9] combines causal self-attention with temporal information to model learning and forgetting. RobustKT [8] decouples cognitive patterns from random factors in response histories.

Each baseline receives historical question or concept IDs, observed responses, and target IDs through its standard input interface. Cognitive-RAG KT additionally uses question text and collaborative representations, together with retrieved evidence and LLM-generated summaries. All methods use the same data splits, history constraints, and test targets.

4.3 Implementation Details

Rasch difficulty, collaborative representations, and graph statistics based on interaction or co-occurrence data are estimated from training interactions and fixed during evaluation. Standardization, clustering, and PCA are also fitted on training data before being applied to test items. A frozen pretrained text encoder processes target question text and public metadata. Missing target features are handled using training-derived priors, OOV or zero representations with missingness indicators, or graph mappings based on an unknown node or public metadata, as appropriate.

We set the final evidence budget to $K = K_2 = 6$ in the main experiments. In the prediction stage, Qwen3-8B remains frozen and only the historical-context and question-context projection networks are trained. We report AUC, ACC, F1, and RMSE, with AUC as the primary metric.

5 Results and Analysis

5.1 Overall Prediction Performance

Table 3 shows that Cognitive-RAG KT achieves the highest AUC on all three datasets. Compared with DenoiseKT, the strongest baseline in AUC, it obtains absolute AUC gains of 0.1248, 0.1158, and 0.0587 on FoundationalASSIST, XES3G5M, and MOOCRadar,

Table 3. Prediction performance on three datasets. The best result is bold, and the best baseline result is underlined.

Method	FoundationalASSIST				XES3G5M				MOOCRadar			
	AUC	ACC	F1	RMSE	AUC	ACC	F1	RMSE	AUC	ACC	F1	RMSE
DKT	0.5053	0.5053	0.5621	0.5085	0.5006	0.5013	0.6125	0.5100	0.4830	0.4932	0.6062	0.5161
SAKT	0.5974	0.5902	0.6667	0.5396	0.5636	0.7011	0.8127	0.4960	0.7090	0.7779	0.8666	0.4103
SAINT	0.6184	0.6226	0.7144	0.4857	0.5692	0.7831	0.8773	0.4211	0.6637	0.7024	0.8107	0.4476
AKT	0.6430	0.6360	0.7195	0.4855	0.6065	0.7503	0.8504	0.4235	0.7357	0.8099	0.8895	0.3716
DIMKT	0.6250	0.6388	0.7414	0.4839	0.5771	0.7607	0.8601	0.4228	0.7385	0.8345	0.9066	0.3518
QIKT	0.6250	0.6378	0.7377	0.4857	0.6049	0.7712	0.8670	0.4154	0.7806	0.8371	0.9081	0.3444
TCKT	0.6011	0.5967	0.6760	0.5130	0.6004	0.7635	0.8613	0.4193	0.7702	0.8220	0.8965	0.3559
SimpleKT	0.5981	0.5625	0.6158	0.5367	0.5726	0.7312	0.8363	0.4431	0.7582	0.8207	0.8953	0.3596
SparseKT	0.6067	0.6067	0.6918	0.5109	0.5810	0.7338	0.8381	0.4407	0.7387	0.8180	0.8941	0.3668
DenoiseKT	0.6557	0.6469	0.7340	0.4694	0.6384	0.7839	0.8768	0.4028	0.7817	0.8391	0.9094	0.3415
RobustKT	0.6252	0.6098	0.6835	0.4903	0.5745	0.7413	0.8448	0.4392	0.7448	0.8297	0.9029	0.3555
KeenKT	0.6246	0.6226	0.7054	0.4824	0.6232	0.7612	0.8577	0.4182	0.7758	0.8067	0.8849	0.3664
Cognitive-RAG KT	**0.7805**	**0.7335**	**0.7998**	**0.4239**	**0.7542**	**0.8046**	**0.8873**	**0.3791**	**0.8404**	**0.8521**	**0.9111**	**0.3322**

respectively. These results support the framework's effectiveness for response prediction on questions involving new KCs under the evaluated setting.

Cognitive-RAG KT also achieves the highest ACC and F1 and the lowest RMSE on all three datasets. Relative to DenoiseKT, RMSE decreases by 0.0455, 0.0237, and 0.0093 on FoundationalASSIST, XES3G5M, and MOOCRadar, respectively. These gains cover both classification metrics and probability prediction error.

5.2 Ablation Study

We evaluate three ablations of retrieval and evidence construction in Table 4. The w/o Retrieval (Recent-K) variant replaces target-conditioned selection with the most recent K admissible interactions and generates its summary from this replacement history. The target information, evidence budget, summary-generation procedure, predictor, and training settings remain unchanged. The w/o Summary variant removes the generated summary text and its derived features $\mathbf{f}_t$ from the historical context. The w/o Fine-Grained Evidence variant replaces the entire historical context x_t^{hist} with a 12-dimensional vector of coarse numerical features summarizing the student's response history and target position. The replaced context includes the retrieved interaction records, relation attributes, and generated summary. The target-question branch is retained. All variants use the same data splits and evaluation targets.

Target-conditioned retrieval yields the largest AUC gain in the component comparison. Relative to Recent-K, the full model improves AUC by 0.0382, 0.0522, and 0.0540 on FoundationalASSIST, XES3G5M, and MOOCRadar, respectively, and also achieves higher ACC and F1 on each dataset. With the evidence budget and downstream predictor fixed, these gains support target-conditioned history selection over recency-based selection.

The full model also outperforms both w/o Summary and w/o Fine-Grained Evidence on AUC, ACC, and F1 across all three datasets. Removing the summary text and its derived features reduces AUC by 0.0019–0.0064. Compared with w/o Fine-Grained

Table 4. Ablation results for retrieval and context construction. Recent-K uses the most recent K admissible interactions. w/o Fine-Grained Evidence replaces the evidence context with coarse numerical features.

Variant	FoundationalASSIST			XES3G5M			MOOCRadar		
	AUC	ACC	F1	AUC	ACC	F1	AUC	ACC	F1
Full	**0.7805**	**0.7335**	**0.7998**	**0.7542**	**0.8046**	**0.8873**	**0.8404**	**0.8521**	**0.9111**
w/o Retrieval (Recent-K)	0.7423	0.7030	0.7728	0.7020	0.7894	0.8791	0.7864	0.8204	0.8958
w/o Summary	0.7786	0.7240	0.7896	0.7517	0.8014	0.8746	0.8340	0.8439	0.9036
w/o Fine-Grained Evidence	0.7731	0.7280	0.7980	0.7521	0.8030	0.8858	0.8394	0.8439	0.9082

Evidence, the full model improves AUC by 0.0074, 0.0021, and 0.0010 on FoundationalASSIST, XES3G5M, and MOOCRadar, respectively. These comparisons support the complete evidence context, including the summary text and its derived features, for response prediction.

5.3 Sensitivity to Retrieval Size

We vary the final evidence budget $K = K_2$ over $\{2, 4, 6, 8, 10\}$ on XES3G5M, keeping the candidate recall budget K_1 and all other settings fixed. As shown in Figure 2, AUC and ACC increase and RMSE decreases as K rises from 2 to 6. At $K = 6$, AUC reaches 0.7542 and RMSE reaches 0.3791, their best values among the evaluated settings. Increasing K to 8 or 10 does not further improve either metric, although ACC peaks at 0.8051 at $K = 8$. With AUC as the primary metric, we use $K = 6$ in the main experiments.

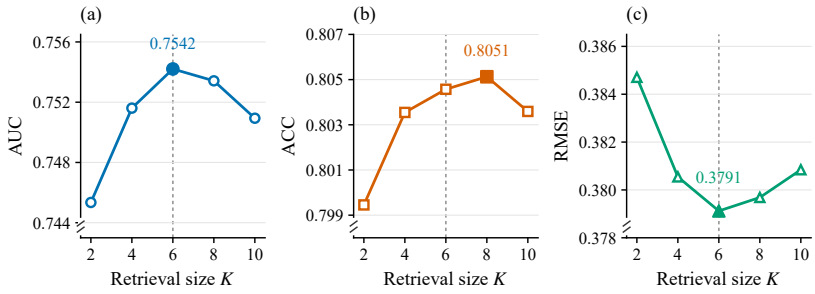


Fig. 2. Sensitivity to retrieval size on XES3G5M: (a) AUC, (b) ACC, and (c) RMSE. Dashed vertical lines indicate $K = 6$, the setting used in the main experiments. Filled markers indicate the best value for each metric. Each panel uses a local vertical-axis range, with a scale-break symbol indicating the omitted interval.

5.4 Case Study

(a) Summary-generation input

<System input: excerpt>
Output one valid JSON object containing the six required fields. Do not output explanations or extra text.

<User input: excerpt>
Generate a structured diagnostic JSON from the target question, performance summary, and selected cognitive evidence.
Use concepts from the target or evidence overlap; cite the input evidence IDs.

Target_ID: Q_1949

Knowledge_concepts:
Periodic problems in basic arrangements; Periodic problems.

Target_text:
101 black and white beads are arranged ... What is the final bead's color, and how many beads have that color?

Performance_summary:
Recent performance is stable and relatively strong; judgment mainly relies on graph-supplement evidence; overall risk is low.

<Selected evidence: excerpts>

evidence_id=1; hist_pid=Q_294; ...
text=... What is the 100th shape? ...
answer=correct; ...

evidence_id=2; hist_pid=Q_301; role=Collaborative supplement; level_diff=-1; answer=correct; support_score=2.2548; ...
text=249 flowers repeat as 5 red, 9 yellow, and 13 green. What is the last flower's color? Which color is most/least common, and by how many? ...

evidence_id=3; answer=correct; ...
text=2, 3, 4, 5, 2, 3, 4, 5, ... What is the 44th number?

evidence_id=4; answer=correct; ...
text=... What is the color of the 65th light? ...

evidence_id=5; answer=correct; ...
text=Flags repeat as 1 red, 2 green, and 3 yellow. How many green flags are among the first 60?

evidence_id=6; answer=correct; ...
text=... What is the color of the 45th bead? ...

(b) Saved summary

<JSON output: translated>

```
{
 "mastered_concepts": [{
 "concept": "Periodic problems in basic
 arrangements",
 "evidence_ids": [1, 2, 3, 4, 5, 6],
 "confidence": "High"
 }],
 "weak_concepts": [],
 "transfer_state": "Stable homogeneous
 transfer",
 "risk_level": "Low",
 "evidence_quality": "High",
 "diagnosis": "Evidence for periodic problems
 is consistent, with good transfer and no
 apparent weaknesses."
}
```

(c) Response prediction

<Prediction input>
Knowledge tracing task: Using the student's current cognitive state and the target requirements, determine whether the response is more likely correct or incorrect.

Student state:
<H1><H2><H3><H4>

Target question:
Knowledge_concepts:Periodic problems in basic arrangements; Periodic problems.
Cognitive level=2; text=101 black and white beads are arranged as:
○●○○○●○○○●○○○ ...
What is the final bead's color, and how many beads have that color?
Exercise answer options: [...]

Target representation:
<Q1><Q2>

Output label (response correctness):
A=correct, B=incorrect. Output one label only.

Label:

<Prediction result>
P(correct) = 0.7663
Observed response: Correct

Translated and abridged excerpts; ... marks omissions. H/Q indicate continuous soft slots.
Seen KC: Periodic problems in basic arrangements. New KC: Periodic problems.
A/B encode response correctness; exercise answer options are omitted from this excerpt.

Fig. 3. Case study of Q_1949 on XES3G5M. (a) Reconstructed summary-generation input excerpts. (b) The saved structured summary. (c) The prediction prompt and result. Text is translated and abridged; ellipses indicate omissions, and H/Q mark continuous soft-slot positions. Gray highlights identify passages discussed in the text. A/B denote response correctness; exercise answer options are omitted from the displayed excerpt.

Figure 3 presents a case from XES3G5M for target question Q_1949. The target involves the training concept "Periodic problems in basic arrangements" and the new concept "Periodic problems". The question presents a repeating arrangement of 101 beads and asks for the final bead's color and the number of beads of that color.

All six selected historical questions involve the training concept, and the student answered each correctly. E1, E3, E4, and E6 concern identifying elements at specified positions, while E5 concerns counting occurrences. E2 combines final-position identification with quantity comparison and has cognitive level 3, compared with level 2 for the target. Together, these records document correct responses on historical position-finding and counting tasks related to the target.

The saved summary in panel (b) was generated from the inputs in panel (a). It cites E1–E6 as evidence for mastery of the shared training concept. It describes the pattern of correct responses on these related questions as "stable homogeneous transfer". Panel (c) shows the prediction prompt, with four history slots H1–H4 before the target question and two target slots Q1–Q2 after it. The correct and incorrect label logits yield a correct-response probability of 0.7663, consistent with the student's observed correct response.

6 Conclusion

We propose Cognitive-RAG KT for predicting student responses to questions involving new KCs through target-conditioned evidence retrieval. It combines multiple relations and balances relevance, coverage, and redundancy to select relevant and complementary historical interactions. Structured evidence attributes and a schema-constrained summary organize these interactions into a context representation. This context and the target-question representation condition a frozen language model through soft slots. On three public datasets, the framework outperforms the evaluated standard KT baselines. Component ablations show higher prediction performance with target-conditioned retrieval than with the most recent K admissible interactions. Future work will explore evidence caching and lightweight predictors to improve inference efficiency and support deployment in intelligent tutoring systems.

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